Auteur: Stan Van Campenhout
Studierichting: Informatica
Oefeningenreeks 4A nr. 57.4

$$\int \frac{1}{\tan x - 1} dx \overset{t=\tan x}{=} \int \frac{1}{t-1} \cdot \frac{1}{1+t^2} dt$$
$$\rightarrow \frac{1}{(t-1)(1+t^2)} = \frac{A}{t-1} + \frac{Bx+C}{1+t^2}$$
$$\Rightarrow A = \frac{1}{1^2+1} = \frac{1}{2}$$
$$\Rightarrow B(i) + C = \frac{1}{i-1} = \frac{i+1}{-2} \Rightarrow B = -\frac{1}{2}, C = -\frac{1}{2}$$
$$= \frac{1}{2} \int \frac{1}{t-1} dt - \frac{1}{2} \int \frac{t+1}{t^2+1} dt$$
$$= \frac{1}{2} |t-1| - \frac{1}{2} \int \frac{t}{t^2+1} dt - \frac{1}{2} \int \frac{1}{t^2+1} dt$$
$$= \frac{1}{2} \ln|t-1| - \frac{1}{4} \ln|t^2+1| - \frac{1}{2} \operatorname{Bgtan} t + c$$
$$\overset{t=\tan x}{=} \frac{1}{2} \ln|\tan x - 1| - \frac{1}{4} \ln|\tan^2 x + 1| - \frac{1}{2} x + c$$

References